

Rice

Application of Organic Manures:

- Apply FYM 12.5 t/ha or compost or green leaf manure @ 6.25 t/ha.
- If green manure is raised @ 50 kg seeds/ha in situ, incorporate it to a depth of 15 cm using a green manure trampler or tractor.
- In the place of green manure, press mud / composted coir pith can also be used.

Stubble Incorporation:

- Apply 22 kg urea / ha at the time of first puddling while incorporating the stubbles of previous crop to compensate immobilization of N by the stubbles.
- This may be done at least 10 days prior to planting of subsequent crop.
- This recommendation is more suitable for double crop wetlands, wherein, the second crop is transplanted in succession with short turn around period.

Application of Biofertilizers :

Raise Azolla as a dual crop by inoculating 500 kg/ha 7 to 10 DAT and then incorporate during weeding for the wet season crop.

Two kg each of biofertilizers viz., Azospirillum, Phosphobacteria (or) Azophos, Silicate solubilizing bacteria (SSB) / Potash bacteria (KRB) with 25kg of FYM and 25kg of sand and broad cast uniformly before transplanting.

500 ml each of biofertilizers viz., Azospirillum, Phosphobacteria (or) Azophos, Silicate solubilizing bacteria (SSB) / Potash bacteria (KRB) with 25kg of FYM and 25kg of sand and broad cast uniformly before transplanting.

Root Dipping :

Prepare the slurry with Azospirillum , Phosphobacteria and potash release bacteria each 1 kg/ha in 40 lit. of water and dip the root portion of the seedlings for 15-30 minutes in bacterial suspension and transplant.

Liquid Azospirillum , Phosphobacteria and potash release bacteria each 250ml/ha in 40 lit. of water and dip the root portion of the seedlings for 15 - 30 minutes in bacterial suspension and transplant.

Root Dipping with Carrier Based Biofertilizers

Zinc Deficient Soils: Along with existing recommended biofertilizers, 2 kg each zinc solubilizing bacterium with 25 kg of FYM and 25 kg of sand and broad cast uniformly before transplanting.

Terminal Water Stress: Foliar spray@ 500 ml/ha of Pink Pigmented Facultative Methylophobus at panicle initiation & flag leaf stages) for mitigation of terminal drought.

Sorghum

Nursery

Application of FYM to the nursery

- Apply 750 kg of FYM or compost for 7.5 cents nursery and apply another 500 kg of compost or FYM for covering the seeds after sowing.
- Spread the manure evenly on the unploughed soil and incorporate by ploughing or apply just before last ploughing.

MAIN FIELD

APPLICATION OF FYM

Spread 12.5 t/ha FYM or composted coir pith along with 10 packets of Azospirillum (2000g/ha) and 10 packets (2000 g/ha) of phosphobacteria or 20 packets of Azophos (4000g/ha) on the unploughed field and incorporate the manure in the soil. Apply well decomposed poultry manure @ 5 t/ha to improve the grain yield as well as physical properties of soils.

APPLICATION OF FERTILIZERS

Transplanted crop

1. Apply NPK fertilizers as per soil test recommendations. If soil test recommendations are not available, adopt a blanket recommendation of 90 N, 45 P₂O₅, 45 K₂O kg/ha.
2. Apply N @ 50:25:25 % at 0, 15 and 30 DAS and full dose of P₂O₅ and K₂O basally before planting
3. In the case of ridge planted crop, open a furrow 5 cm deep on the side of the ridge at two thirds the distance from the top of the ridge and place the fertilizer mixture along the furrow and cover with soil upto 2 cm.
4. Soil application of Azospirillum at 10 packets (2 kg/ha) and 10 packets (2000g/ha) of phosphobacteria or 20 packets of Azophos (4000g/ha) after mixing with 25 kg of FYM + 25 kg of soil may be carried out before sowing/planting.

Direct sown crop

- Apply NPK fertilizers as per soil test recommendations as far as possible. If soil test recommendations are not available, adopt a blanket recommendation of 90 N, 45 P₂O₅, 45 K₂O kg/ha.
- Apply N @ 50:25:25 % at 0, 15 and 30 DAS and full dose of P₂O₅ and K₂O basally before sowing and if basal application is not possible the same could be top dressed within 24 hours.
- In the case of bed planted crop, mark lines to a depth of 5 cm and 45 cm apart. Place the fertilizer mixture at the depth of 5 cm along the lines. Cover the lines upto 2 cm from the top before sowing.
- In the case of sorghum raised as a mixed crop with a pulse crop (Blackgram, Greengram or Cowpea) open furrows 30 cm apart to a depth of 5 cm.
- Apply fertilizer mixture in two lines in which sorghum is to be raised and cover upto 2 cm.

- Skip the third row in which the pulse crop is to be raised and place fertilizer mixture in the next two rows and cover upto 2 cm with soil.
- Application of bio-fertilizers: When Azospirillum is used apply only 75% of recommended N for irrigated sorghum.
- Soil test based fertilizer recommendation may be adopted in Western and North Western Zone viz., Alfisol, Inceptisol and Vertisol for prescribing fertilizer doses for specified yield targets.

APPLICATION OF MICRONUTRIENT MIXTURE

Transplanted Crop

- Mix 12.5 kg/ha of micronutrient mixture formulated by the Department of Agriculture, Tamil Nadu with enough sand to make a total quantity of 50 kg and apply the mixture over the furrows and on top one third of the ridges.
- If micronutrient mixture is not available, mix 25 kg of zinc sulphate with sand to make a total quantity of 50 kg and apply on the furrows and on the top one third of the ridges.

Direct Sown Crop

- Mix 12.5 kg of micronutrient mixture formulated by the Department of Agriculture, Tamil Nadu with enough sand to make a total quantity of 50 kg.
- Spread the mixture evenly on the beds.
- Basal application of 25 kg ZnSO₄/ha-1 for the deficient soils or 12.5 kg Zn SO₄ ha-1 + FYM for the Zn deficient soils
- Basal application of FeSO₄, 50 kg/ha along with 12.5 t/ha FYM for iron deficient soils.

Direct sown crop

- Spray only if micronutrient mixture is not applied.
- Apply in case of iron deficiency.
- If soil is calcareous

RATOON SORGHUM CROP

APPLICATION OF FERTILIZERS

- Apply 100 kg N/ha in two split doses.
- Apply the first dose on 15th day after cutting and the second on 45th day after cutting.
- Apply 50 kg P₂O₅/ha along with the application of N on 45th day.

RAINFED SORGHUM

Test the soil and apply fertilizers based on soil test recommendations.

FYM application should be done @ 12.5 t / ha and well incorporated at the time of ploughing. Chisel the soils having hard pan formation at shallow depths with chisel plough at 0.5 M interval, first in one direction and then in the direction perpendicular to the previous one once in three years. Apply 12.5 t FYM or composted Coir pith/ha besides chiseling to get an additional yield of about 30% over control.

- Treat the seeds with three packets of azospirillum (600 g) and 3 packets of phosphobacteria or 6 packets of Azophos (1200 g/ha). In the main field, apply 10 packets of azospirillum 2000 g/ha and 10 packets (2000g/ha) of phosphobacteria or 20 packets of Azophos (4000 g/ha) with phosphobacteria 2 kg with 25 kg FYM + 25 kg soil.

FERTILIZER

Apply 12.5 t/ha of Composted Coir pith + NPK at 40:20:0; Apply enriched FYM 750 kg/ha. The recommended dose of 40 kg N and 20 kg P₂O₅/ha for rainfed sorghum can be halved if FYM @ 5 t/ ha is applied.

Ragi

Ragi (*Eleusine coracana*): Irrigated

Preparation of Main Field

1. Application of FYM or Compost

Spread 12.5 t/ha of FYM or compost or composted coir pith evenly on the unploughed field and then plough and incorporate in the soil.

NOTE: Do not spread and leave the manure uncovered in the field as nutrients will be lost.

2. APPLICATION OF FERTILIZERS

- Soil test based fertilizer recommendations may be adopted in Western and North Western Zones in soils of Vertisols, Alfisols, Inceptisols and Entisols for prescribing the fertilizer doses for specified yield targets. In soils having high intensive cropping system viz., Ragi-Maize-Cowpea, having high soil available K potassium need not be applied. Apply NPK fertilizers as per soil test recommendation as far as possible. If soil test recommendation is not available, adopt a blanket recommendation of 60 kg N, 30 kg P₂O₅ and 30 kg K₂O per ha.
- Apply half the dose of N and full dose of N and full dose of P₂O₅ and K₂O and basally before planting.
- Broadcast the fertilizer mixture over the field before the last ploughing and incorporate into the soil by working a country plough.
- Apply 10 packets/ha (2000 g) of azospirillum and 10 packets (2000 g/ha) of Phosphorous solubilizing bacteria or 20 packets of Azophos (4000 g/ha) after mixing with 25 kg of soil and 25 kg FYM before transplanting.

3. APPLICATION OF MICRONUTRIENT MIXTURE

- Mix 12.5 kg of micronutrient mixture formulated by the Department of Agriculture, Tamil Nadu with enough sand to make a total quantity of 50 kg/ha.
- Apply the mixture evenly on the beds.
- Do not incorporate the mixture in the soil.

4. MANAGEMENT OF MAIN FIELD

Root dipping with Azospirillum: Prepare slurry with 5 packets (1000 g/ha) of Azospirillum and 5 packets (1000g/ha) of Phosphobacteria or 10 packets of Azophos (2000 g/ha) in 40 litres of water and dip the root portion of the seedlings in the solution for 15-30 minutes and transplant.

RAGI : RAINFED

1. Manuring and fertilization

Finger millet responds well to fertilizer application especially to N and P.

The recommended doses of fertilizers vary from state to state for rainfed crop. Apply recommended dose of 40:20:20 N:P:K kg/ha was applied. With judicious application of farmyard manure, inorganic fertilizer efficiency is enhanced.

Entire P₂O₅ and K₂O are to be applied at sowing, whereas nitrogen is to be applied in two or three split doses depending upon moisture availability.

In areas of good rainfall and moisture availability, 50% of recommended nitrogen is to be applied at sowing and the remaining 50% in two equal splits at 25-30 and 40-45 days after sowing.

In areas of uncertain rainfall, 50% at sowing and the remaining 50% around 35 days after sowing is recommended.

2. Bio-fertilizers

Treating seeds with *Azospirillum brasilense* (N fixing bacterium) and *Aspergillus awamori* (P solubilizing fungus) @ 25 g/kg seed is beneficial.

In case seeds are to be treated with seed dressing chemicals, treat the seeds first with seed dressing chemicals and then with bio-fertilizers at the time of sowing.

3. Procedures for inoculating seeds with biofertilizers

1. Bio-fertilizer culture specific to the crop is to be used @ 25 g per kg of seed.
2. Sticker solution is necessary for effective seed inoculation. This can be prepared by dissolving 25 g jaggery or sugar in 250 ml water and boiling for 5 minutes. The solution thus prepared is cooled.
3. Smear the seeds well using the required quantity of sticker solution. Then add culture to the seeds and mix thoroughly so as to get a fine coating of culture on the seed.
4. The culture-coated seeds is to be dried well in shade to avoid clumping of seeds.
5. Use the inoculated seeds for sowing.

Cumbu

Cumbu (*Pennisetum glaucum* (L) R. Br.)

Nursery

Application of FYM

Apply 750 kg of FYM or compost and incorporate by ploughing. Cover the seeds with 500 kg of FYM.

PREPARATION OF MAIN FIELD

APPLICATION OF FYM OR COMPOST

Spread 12.5 t/ha of FYM or compost or composted coir pith uniformly on unploughed soil. Incorporate the manure by working the country plough and apply Azospirillum to the soil @ 10 packets per ha (2000 g) and 10 packets (2000g) of phosphobacteria (or) 20 packets (4000g) of azophos with 25kg of soil and 25 kg of FYM.

APPLICATION OF FERTILIZERS

Apply NPK fertilizers as per soil test recommendations as far as possible. If soil test recommendation is not available follow the blanket recommendation of 70:35:35 kg N, P₂O₅, K₂O/ ha for all varieties. For hybrids, apply 80 kg N, 40 kg P₂O₅ and 40 kg K₂O per ha. Apply the recommended N in three splits as 25:50:25 per cent at 0,15 and 30 DAS and full dose of phosphorus and potassium basally. Combined application of azospirillum and phosphobacteria or azophos along with 75 per cent of the recommended level of N and P is recommended for rainfed conditions.

Method of application: For transplanted crop, open a furrow more than 5 cm deep on the side of the ridge (1/3 distance from the bottom), place the fertilizer and cover. For the direct sown crop, mark the lines more than 5 cm deep 45 cm apart in the beds. Place the fertilizer below 5 cm depth and cover upto 2 cm from the top before sowing. In the case of intercropping with pulses, mark lines more than 5 cm deep 30 cm apart in the beds. Apply fertilizer only in the rows in which cumbu is to be sown and cover upto 2 cm. When azospirillum inoculant is used for seeds, seedlings use only 50 kg N/ha for variety, 60 kg N/ha for hybrid, as soil application in other words, reduce 25% N of soil test recommendations.

APPLICATION OF MICRONUTRIENT MIXTURE

Apply 12.5 kg/ha of micronutrient mixture formulated by the Department of Agriculture. Mix the mixture with enough sand to make 50 kg and apply on the surface just before planting/after sowing and cover the seeds. Broadcast the mixture on the surface to seed line. If micronutrient mixture is not available apply 25 kg of zinc sulphate per ha. Mix the chemical with enough sand to make 50 kg and apply as above.

MANAGEMENT OF MAIN FIELD

1. Root dipping with bio-fertilizers: Prepare the slurry with 5 packets (1000 g)/ha of Azospirillum inoculant and 5 packets (1000g/ha) of phosphobacteria or 10 packets of azophos (2000g/ha) in 40 lit. of water and dip the roots of the seedlings 15 - 30 minutes before planting.

TOP DRESSING OF FERTILIZERS

1. Top dress the nitrogen at 15 and 30 days after transplanting or direct sowing.
2. In transplanted crop, open a furrow 5 cm deep with a stick or hoe at the bottom of the furrow, place the fertilizer and cover.
3. In the case of direct sown crop apply the fertilizer in band. If intercropped with pulses apply the fertilizer to cumbu crop only.
4. After the application of fertilizer, irrigate the crop.

Varagu

1. Apply 12.5 t/ha of FYM/Compost basally on the unploughed field and then plough and incorporate
2. Apply a blanket recommendation of 44: 22 kg NP/ha as inorganic fertilizer
3. Apply Azospirillum + Phosphobacteria 2 kg each/ha or Azophos 4 kg /ha basally.

Sugarcane

Sugarcane (*Saccharum officinarum*)

Basal application of organic manures:

1. Apply FYM at 12.5 t/ha or compost 25 t/ha or filter press mud at 37.5 t/ha before the last ploughing under gardenland conditions.
2. In wetlands this may be applied along the furrows and incorporated well.

Basal Application of Fertilizer

1. If soil test is not done, follow blanket recommendation of NPK @ 300:100:200 kg/ha Apply super phosphate (625 kg/ha) along the furrows and incorporate with hand hoe.
2. Apply 37.5 kg Zinc sulphate/ha and 100 kg Ferrous sulphate/ha to zinc and iron deficient soils.
3. Application of sulphur in the form of Gypsum @ 500 kg /ha to sulphur deficient soils to increase the cane yield and juice quality.

Top Dressing with Fertilizers

a. Soil application

Apply 275 kg of nitrogen and 112.5 kg of K₂O/ha in three equal splits at 30, 60 and 90 days in coastal and flow irrigated belts (assured water supply areas).

In the case of lift irrigation belt, apply 225 kg of nitrogen and 112.5 kg of K₂O/ha in three equal splits at 30, 60 and 90 days (water scarcity areas). For jaggery areas, apply 175 kg of nitrogen and 112.5 kg of K₂O/ha in three equal splits on 30, 60 and 90 days.

Nitrogen Saving

- **Neem Cake Blended Urea:** Apply 67.5 kg of N/ha + 27.5 kg of Neem Cake at 30 days and repeat on 60th and 90th days.
Note: Neem cake blending: Powder the required quantity of neem cake and mix it with urea thoroughly and keep it for 24 hours. Thus, 75 kg of nitrogen/ha can be saved by this method.
- **Azospirillum:** Mix 12 packets (2400 g)/ha of Azospirillum inoculant or TNAU Biofert –1 with 25 kg of FYM and 25 kg soil and apply near the clumps on 30th day of planting. Repeat the same on 60th day with another 12 packets (2400 gm). Repeat the above on the other side of the crop row on the 90th day (for lift irrigated belt).
- **Band placement:** Open deep furrows of 15 cm depth with hand hoes and place the fertilisers in the form of band and cover it properly.

- **Subsurface application:** Application of 255 kg of Nitrogen in the form of urea along with potash at 15 cm depth by the side of the cane clump will result in the saving of 20 kg N/ha without any yield reduction.

Importance of Balanced Nutrition

The soil fertility has declined in many sugarcane growing areas of the state due to improper and some times, distorted fertilizer schedules adopted over the years under intensive cultivation of the crop. Hence balanced application of fertilizer based on soil test values and crop requirement is essential.

Micro nutrient fertilizers :

- (a) Zinc deficient soils : Basal application of 37.5 kg/ha of zinc sulphate.
(b) Sugarcane crop with zinc deficiency symptoms: foliar spray of 0.5% zinc sulphate with 1% urea at 15 days interval till deficiency symptoms disappear.
- (a) Iron deficient soils: Basal application of 100 kg/ha of ferrous sulphate.
(b) Sugarcane with Iron deficiency symptoms: foliar spray of 1% ferrous sulphate with 1% urea at 15 days interval till deficiency symptoms disappear.
- Soil application of CuSO₄ @ 5 kg/ha in copper deficient soils. Alternatively foliar spray of 0.2% CuSO₄ twice during early stage of crop growth.

Common Micronutrient mixture : To provide all micronutrients to sugarcane, 50 kg /ha of micronutrient mixture containing 20 kg Ferrous sulphate, 10 kg Manganese sulphate, 10 kg Zinc sulphate, 5 kg of Copper sulphate, 5 kg of Borax mixed with 100 kg of well decomposed FYM, can be recommended as soil application prior to planting.

(Or) Application of TNAU MN mixture @ 50 kg/ha as EFYM for higher cane yield.

Recommended dosage of macro and micronutrients

Macronutrients

- Sugarcane – plant crop (meant for sugar mills) 300:100:200 kg N, P₂O₅ and K₂O per ha
- Sugarcane – Ratoon crop (meant for sugar mills)
- 300 + 25% extra N : 100 : 200 kg N, P₂O₅ and K₂O per ha
- Sugarcane for jaggery manufacture (plant as well as ratoon crop) 225 : 62.5 : 112.5 kg N, P₂O₅ and K₂O per ha.

BIOFERTILIZER FOR SUGARCANE

Azospirillum is the common biofertilizer recommended for N nutrition which could colonize the roots of sugarcane and fix atmospheric nitrogen to the tune of about 50 to 75 kg nitrogen per ha per year. Recently, another endophytic nitrogen fixing bacterium, *Gluconacetobacter diazotrophicus* isolated from sugarcane can able to fix more nitrogen than *Azospirillum*. It colonizes throughout the sugarcane and increases the total N content. In soil, it can also colonize the roots and able to solubilize the phosphate, iron and Zn. It can also enhance the crop growth, yield of sugarcane and sugar content of the juice. Since it is more efficient than *Azospirillum*, this new organism was test- verified in various centres and released as new biofertilizer *Gluconacetobacter diazotrophicus* TNAU Biofert-I. Phosphobacteria as P solubiliser are recommended for sugarcane crop.

Sett treatment with *Gluconacetobacter diazotrophicus*

Before planting the sugarcane setts can be treated with ten packets (2 kg) per ha of *Gluconacetobacter diazotrophicus* prepared as slurry with 250 L of water.

Soil application *Gluconacetobacter diazotrophicus*

Twelve packets (2.4 kg) per ha is recommended for soil application each at 30th, 60th and 90th day after planting under irrigated condition.

Same method of application can be followed for Phosphobacteria.

- If basal application is not followed apply the same with 30th day, 60th day and 90th day after planting and copiously irrigate the field.
- Biofertilizer treatment should be done just before planting. Immediately plant/ Irrigate after biofertilizer application.
- Do not mix biofertilizer along with chemical fertilizer.
- Reduces 25% of the recommended N to reap the benefits of biofertilizer application

Management of the Crop

1. 25% additional N application on 5-7 days after ratooning.
2. Spray Ferrous sulphate at 2.5 kg/ha on the 15th day. If chlorotic condition persists, repeat twice further at 15 days interval. Add urea 2.5 kg/ha in the last spray.
3. First top dressing on 25th day, 2nd on 45th to 50th day.
4. Final manuring on 70th to 75th day.

Short Crop (Nursery Crop)

Apply 50 kg of urea as top dressing additionally before one month of cutting the seed cane.

Sugarbeet

Tropical Sugarbeet

Manures and Fertilizers

S.No	Manures and Fertilizers	Basal Application	Top dressing
1	Manures	12.5 tonnes /ha	-
2	Biofertilizers Azospirillum Phosphobacteria	2 kg /acre (10 pockets) 2 kg /acre (10 pockets)	-

3.	FertilizersNitrogen (150)	75kg /ha	37.5 kg / ha each at 25 & 50 DAS
	Phosphorus (75)	75kg /ha	-
	Potassium (75)	75kg /ha	-

Right nutrition

- Balanced application of organic and inorganic fertilizers
- FYM 12.5 t/ha and basal application of 2 kg of Azospirillum and 2 kg of phosphobacteria
- Based on the soil test value, inorganic fertilizer has to be applied. In the absence of soil test value, blanket recommendation of 150:75:75 NPK kg/ha
- Stages of application of fertilizer: Basal 50% N, full P and full K.
- Remaining 25% N on 20 to 25 days after sowing and 25% N on 40 to 45 days after sowing.
- Timely and need based, placement of fertilizer and earthing up
- Excess N should be avoided

Fodder Maize

Fodder Maize

Application of FYM

Apply and spread FYM or compost at 25 t/ha on unploughed field along with 10 packets of Azospirillum (2000 g) and 10 packets of Phosphobacteria (2000g) inoculum or 20packets of Azophos (4000g)and incorporate the manure into the soil during ploughing.

Application of Fertilizers

Apply NPK fertilizers as per soil test recommendation as far as possible. If soil testing is not done, follow blanket recommendation of 30 : 40 : 20 kg N, P₂O₅ and K₂O / ha. Apply 30 kg N/ha at 30 days after sowing as top dressing.

Sowing

- Treat the seeds with 3 packets (600 g) Azospirillum inoculant and 3 packets (600 g) of phosphobacteria or 6 packets of Azophos (1200g)before sowing.

Seed Production

Fertilizer

- Apply NPK @ 175 : 90 : 90 kg ha⁻¹ + 25 kg ZnSO₄ ha⁻¹ as basal application.

Soundal

Leucaena – Soundal (*Leucaena leucocephala*)

Application of FYM

Apply and spread 25 t/ha of FYM or compost

Application of Fertilizer

- Apply NPK fertilizers as per soil test recommendations as far as possible. If soil testing is not done, follow the blanket recommendation of 10:60:30 kg NPK/ha.
- Apply full dose of NPK basally before sowing.

FERTIGATION :

Water saving, yield and profit under drip and drip fertigation systems

Crops	Water Saving (%)	Yield (t/ha)			Profit (Rs/ha)		
		Conventional	Drip	Drip+ Fertgn	Conventional	Drip	Drip + Fertgn
Banana	35	26	30	37	81000	98000	120000
Sugarcane	29	120	160	207	30000	47000	68000
Tomato	32	45	56	65	56000	77000	95000

Fertilizers commonly used in fertigation

Name	N – P ₂ O ₅ – K ₂ O content	Solubility (g/l) at 20 C
Ammonium nitrate	34-0-0	1830
Ammonium sulphate	21-0-0	760
Urea	46-0-0	1100
Monoammonium phosphate	12-61-0	282
Diammonium phosphate	18-46-0	575
Potassium chloride	0-0-60	347
Potassium nitrate	13-0-44	316
Potassium sulphate	0-0-50	110
Monopotassium phosphate	0-52-34	230
Phosphoric acid	0-52-0	457

Specialty water soluble fertilizers

Name	N %	P ₂ O ₅ %	K ₂ O %
Polyfeed	19	19	19
Polyfeed	20	20	20
Polyfeed	11	42	11
Polyfeed	16	8	24

Polyfeed	19	19	19
Polyfeed	15	15	30
MAP	12	61	0
Multi-K	13	0	46
MKP	0	52	34
SOP	0	0	50

N fertigation

Urea is well suited for injection in micro irrigation system. It is highly soluble and dissolves in non-ionic form, so that it does not react with other substances in the water. Also urea does not cause precipitation problems. Urea, ammonium nitrate, ammonium sulphate, calcium ammonium sulphate, calcium ammonium nitrate are used as nitrogenous fertilizers in drip fertigation.

P fertigation

Application of phosphorus to irrigation water may cause precipitation of phosphate salts. Phosphoric acid and mono ammonium phosphate appears to be more suitable for fertigation.

K fertigation

Application of K fertilizer does not cause any precipitation of salts. Potassium nitrate, Potassium chloride, Potassium sulphate and mono potassium phosphate are used in drip fertigation.

Micro nutrients

Fe, Mn, Zn, Cu, B, Mo could be used as micro nutrients in drip fertigation.

Fertigation equipments

Three main groups of equipments used in drip system are :

- Ventury
- Fertilizer tank
- Fertilizer pump

Ventury

Construction in the main water flow pipe causes a pressure difference (Vacuum) which is sufficient to suck fertilizer solution from an open container into the water flow. It is very easy to handle and it is affordable even by small farmers. This equipment is most suitable for smaller area.

Fertilizer tank

A tank containing fertilizer solution is connected to the irrigation pipe at the supply point. Part of the irrigation water is diverted through the tank diluting the nutrient solution and returning to the main supply pipe. The concentration of fertilizer in the tank thus becomes gradually reduced.

Fertilizer pump

The fertilizer pump is a standard component of the control head. The fertilizer solution is held in non-pressurised tank and it can be injected into the irrigation water at any desired ratio. Therefore the fertilizer availability to each plants is maintained properly.

Cost of fertigation equipments

Sl.No.	Fertigation devices	Cost (Rs.)
1.	Ventury type	1200
2.	Fertilizer Tank	3000
3.	Injectors	12000

Economics of drip irrigation system

The initial investment in drip irrigation system is mainly depends upon the spacing of crops. The initial cost will be almost 20-25 thousand rupees per hectare for wider spacing crops such as coconut, mango, grapes and for orchard crops. The initial cost is approximately 50-70 thousand rupees per hectare for close spacing crops such as sugarcane, banana, papaya, mulberry, turmeric, tapioca, vegetables and flower crops.

Fertigation Schedule - Maize

Fertigation schedule for Hybrid maize with Water Soluble Fertilizers at (75 % RDF)

Stage (days)	Duration (days)	Fertilizer form	Fertilizer grade			Dose/ha/day	Total Qty (kg/ha)	Nutrients (kg/ha)		
			N	P	K			N	P	K
6 to 25	20	MAP	12	61	0	2.813	56.25	6.75	34.31	0.00
	20	Urea	46	0	0	0.938	18.75	8.63	0.00	0.00
20-60	35	Poly feed	19	19	19	2.143	75.00	14.25	14.25	14.25
	35	Multi-K	13	0	45	1.500	52.50	6.83	0.00	23.63
	35	Urea	46	0	0	2.143	75.00	34.50	0.00	0.00
61-75	15	Poly feed	19	19	19	2.750	41.25	7.84	7.84	7.84
	15	Multi-K	13	0	45	1.600	24.00	3.12	0.00	10.80
	15	Urea	46	0	0	4.500	67.50	31.05	0.00	0.00
								1112.96	56.40	56.51

Fertigation schedule for Hybrid maize with normal fertilizers (100% RDF)

Stage (days)	Duration (days)	Fertilizer form	Fertilizer grade			Dose/ha/day	Total Qty (kg/ha)	Nutrients (kg/ha)		
			N	P	K			N	P	K
6 to 25	20	DAP	18	46	0	5.00	100	18.0	46.0	0.0
	20	Urea	46	0	0	2.50	50	23.0	0.0	0.0
20-60	35	DAP	18	46	0	1.86	65	11.7	29.9	0.0
	35	Urea	46	0	0	4.29	150	69.0	0.0	0.0
	35	MOP	0	0	60	2.14	75	0.0	0.0	45.0
61-75	15	Urea	46	0	0	4.13	62	28.5	0.0	0.0
	15	MOP	0	0	60	3.33	50	0.0	0.0	30.0
								150.2	75.9	75.0

Fertigation Schedule - Red Gram

Fertigation schedule for redgram at 100 % RDF through water soluble fertilizers

Stage	Duration (Days)	Application time (DAS)	Quantity of nutrients - fertilizer grade (kg ha ⁻¹)		
			Urea	MAP	SOP
Seedling	1-30	10, 17 and 24	2.32	32.78	0
Vegetative	31-90	31, 38, 45, 52, 59, 66, 73, 80 and 87	9.87	24.58	12.50
Flowering	91-120	94, 101, 108 and 115	9.87	24.58	20.00
Pod development	121-140	122, 129 and 136	10.85	0	17.50
		Total	32.91	81.94	50.00

Fertigation schedule for redgram at 100 % RDF through conventional fertilizers

Stage	Duration (Days)	Application time (DAS)	Quantity of nutrients - fertilizer grade (kg ha ⁻¹)		
			Urea	MOP	
Seedling	1-30	10, 17 and 24	10.85	0	

Vegetative	31-90	31, 38, 45, 52, 59, 66, 73, 80 and 87	16.28	10.38	
Flowering	91-120	94, 101, 108 and 115	16.28	16.60	
Pod development	121-140	122, 129 and 136	10.85	14.53	
		Total	54.30	41.50	

Fertilizers suitable for fertigation

Fertilizers	N:P:K	Fertilizers	N:P:K
Urea	46 - 0 - 0	Potassium nitrate	13 - 0 - 46
Ammonium nitrate	34 - 0 - 0	MAP	12 - 61 - 0
Ammonium sulphate	21 - 0 - 0	Potassium chloride	0 - 0 - 60
Calcium nitrate	16 - 0 - 0	Potassium nitrate	13 - 0 - 46
Magnesium nitrate	11 - 0 - 0	Potassium sulphate	0 - 0 - 50
Urea ammonium nitrate	32 - 0 - 0	NPK (Polyfeed)	19 - 19 - 19
			20 - 20 - 20

Fertigation Schedule - Sugarcane

Fertigation schedule for sugarcane

Sl.no	Duration (Days)	Fertilizer dose (kg/ha)	
		Urea	Potash
1	15	50	10
2	30	50	10
3	45	50	10
4	60	50	10
5	75	50	10
6	90	50	10
7	105	50	10
8	120	40	15
9	135	40	15
10	150	40	15
11	165	40	15

12	180	30	20
13	195	30	20
14	210	30	20
	Total	600	190

TIPS FOR BETTER FERTILIZER USE EFFICIENCY

1. The fertiliser scheduling must be based on soil test.
2. Selection of fertilisers should be done according to the soil reaction *viz.*, acidic fertiliser for alkaline soils and basic fertilisers for acidic soil reactions.
3. Surface application through broadcasting should not be adopted but the fertiliser should be placed about 3-4 cms by the side or below the seed. This discourages weed growth also.

The phosphatic and potassic fertilisers should be basal placed, because their poor mobility restricts them to the place of application. Therefore, they must be placed in the root zone.

4. Home mixing of fertilisers should be in accordance with the fertilizer-mixing guide and such fertiliser mixture must be applied as soon as possible.
5. In case of heavy soil type, half of the nitrogenous fertilisers should be basal placed and rest should be top-dressed in one split only.
6. But in case of light soils, nitrogen should be applied in three equal splits *i.e* 1/3 as basal, 1/3 after 30 days of sowing and the balance 1/3 about 50-60 days after sowing.
7. For at least a week, flooding with too deep water or poor drainage should be avoided after application of the fertilisers.
8. Top dressing should be done after draining out the water and weeding so that the loss of nutrient is minimum. Paddy fields, used for transplanting, should be puddled and fertilisers should be applied at the time of puddling. This will help fertilisers to penetrate and get stored in the soil.
9. The acidic soils should be treated with liming materials as and when required.
10. Deep placement of fertiliser, along with foliar feeding of nitrogen (*i.e.*, urea) through spraying of nitrogenous fertiliser in place of top dressing should be done in case of dry lands.

- 11. Addition of organic manures or green manuring should be done at least once in 3-5 years. Weed growth should not be permitted in cropped areas, during any part of the year.**
- 12. In case of flooded fields or calcareous soils, use of slow release nitrogenous fertilisers like sulphur coated urea, urea super granules, neem coated or neem blended ureas should be used so that loss of nitrogen can be minimised.**
- 13. Mud bolls, contain urea and should be used in case of deepwater crops because they help in proper placement and also reduce the loss of nitrogen from the field.**
- 14. Appropriate plant protection measures and proper tillage practices should be adopted so that plants remain healthy and absorb the applied nutrients from the field.**